ACI Notes – Part 6

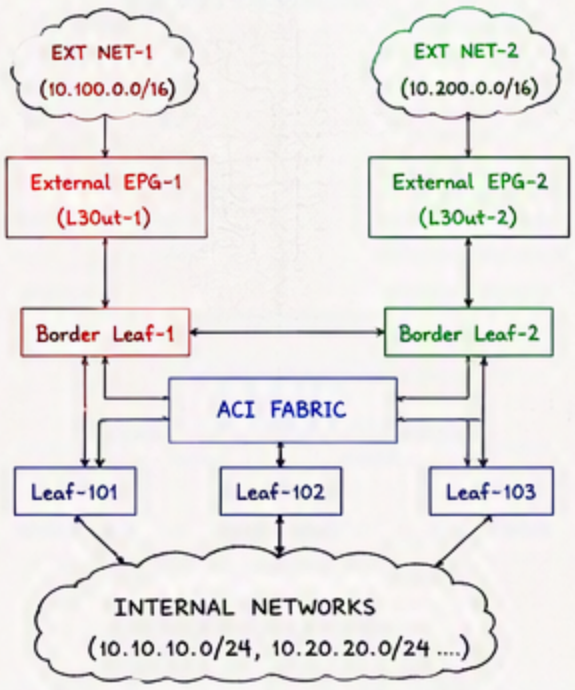
ACI Transit Routing

Routing Between External Networks Through ACI

1. What is Transit

- Routing traffic between two external networks using ACI as the transit device.
- ACI does NOT own the routes. It only forwards traffic.
- Requires TWO L3Outs and appropriate contracts.
- Common in Data Center Interconnect, MPLS ↔ Internet, Partner ↔ DC, etc.

2. Basic Transit Topology



3. How Traffic Flows (EXT-1 → EXT-2)

1. Host in EXT-1 sends traffic to EXT-2.
2. Traffic enters ACI via L3Out-1 / Border Leaf-1.
3. Destination matches External EPG-2.
4. Contract check: EPG-1 (Consumer) → EPG-2 (Provider)
5. Route lookup in ACI VRF.
6. Next-hop = L3Out-2 (Border Leaf-2).
7. Traffic forwarded inside fabric to Border Leaf-2.
8. Traffic exits via L3Out-2 to EXT-2.

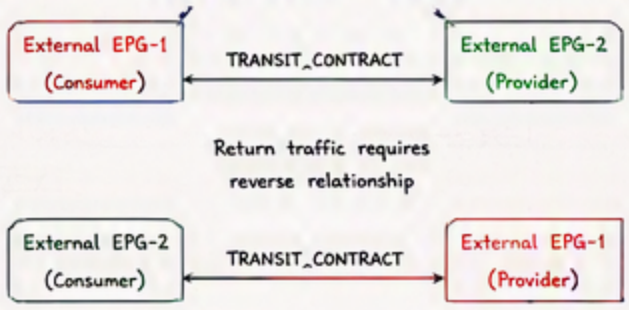
ACI is just a router in between !

4. Requirements

- Two L3Outs (one for each external network).
- External EPG for each L3Out.
- EPG-1 consumes, EPG-2 provides contract (and vice-versa for return traffic).
- Reachability of both external networks inside the VRF (usually static or BGP).
- Return route must exist on remote networks.

5. Contract Model for Transit

Example: EXT-1 → EXT-2

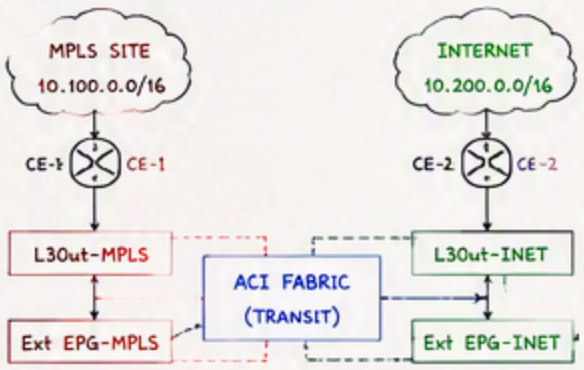


6. Routes Inside ACI VRF

- Route to EXT-1 networks via L3Out-1.
- Route to EXT-2 networks via L3Out-2.
- These can be:
  - Static Routes
  - BGP learned
  - Route Leaking

ACI does NOT learn routes from one external router and advertise them to another by default.

7. Example Scenario



- Traffic from MPLS SITE can reach INTERNET through ACI and vice-versa.

8. Important Considerations

- ✓ ACI does not modify packets (no NAT)
- ✓ ACI only forwards based on routing + policy.
- ✓ MTU must be consistent across L3Out links.
- ✓ Asymmetric routing can cause session issues.
- ✓ Proper route control and contracts are critical.
- ✓ Return path must be available. (Otherwise traffic will not flow back)

9. Route Control in Transit Design

- Use Route Control Profiles on both L3Outs.
- Import only required prefixes.
- Export only required prefixes.
- Prevent leaking unnecessary routes.

Example:

Import: 10.100.0.0/16 (from MPLS)

Export: 10.200.0.0/16 (to MPLS)

10. Types of Transit Routing

1) L3Out-to-L3Out Transit (Most Common)
   Ext-Net-1 ⟷ ACI ⟷ Ext-Net-2
2) VRF-to-External Transit
   VRF-A ⟷ ACI ⟷ Ext-Net
3) VRF-to-VRF via External (Less Common)
   VRF-A ⟷ Ext-Net ⟷ VRF-B

11. External Subnet Flags (Under External EPG)

Import Route Control Subnet (Into ACI)
- Allows prefixes from this subnet to be imported into ACI according to RCP.

Use when ACI needs to learn routes from outside.

Export Route Control Subnet (From ACI)
- Allows prefixes inside ACI to be advertised to this external subnet according to RCP.

Shared Route Control Subnet (Shared Between VRFs / L3Outs)
- Allows prefixes to be shared between VRFs or between L3Outs using same route control.

12. Transit Routing Checklist

1. Both L3Outs are up and BGP is established.
2. External EPGs created for both L3Outs.
3. Contracts configured in both directions.
4. Routes to external networks exist inside ACI VRF.
5. Route Control Profiles correct (Import/Export).
6. No overlapping routes between external networks.
7. MTU consistent end-to-end.
8. Return path available and symmetric routing preferred.

13. Common Issues & Causes

Issue	Possible Cause
No communication between external networks	Missing contract / wrong direction
Route present on BL but not on other Leafs	RCP blocking redistribution / Type-5 not advertised
Traffic one-way	Missing return route or asymmetric routing
Intermittent connectivity	MTU mismatch / PBR issue
BGP up but no routes	Import RCP denying prefixes

14. Verification Commands (Leaf / Border Leaf)

- show ip route vrf <vrf>
- show ip bgp l2vpn evpn route-type 5
- show bgp ipv4 unicast summary
- show bgp l2vpn evpn summary
- show L3out
- show L3out route-control
- show route-map <name>
- show zoning-rule

Always check both routing + policy when troubleshooting!

15. Key Takeaways

- ACI can route between external networks but does not own them.
- Requires two L3Outs + contracts in both directions.
- Route Control Profiles decide what comes in and what goes out.
- ACI forwards based on routing + policy (contracts).
- Return path is critical for successful transit routing.

16. Memory Aid

Think of ACI as a HIGHWAY

L3Out-1 = Entry Ramp
L3Out-2 = Exit Ramp
ACI Fabric = Highway
External EPGs = Signs
Contracts = Traffic Rules
RCP = Toll Gate (Route Control)

Route Types (EVPN)

Type-2 = Endpoint (Host) MAC + IP

Type-5 = Prefix (Network)

Golden Rule for Transit Routing

If traffic does not flow, check in this order:

1. Route
2. Contract
3. Route Control
4. Return Path